

Robotics

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This is a set of notes covering interesting concepts I find while studying the Modern Robotics textbook. I will try to bring analytic theory whenever applicable.

Textbooks Used to Develop these Notes:

1. *Modern Robotics* - Kevin Lynch, Frank Park
2. *Calculus on Manifolds* - Michael Spivak
3. *Mathematical Analysis* - Tom Apostol

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Holonomic Constraints

1.1 Basic Multivariable Analysis

Note: The following notes may not be entirely rigorous! I have attempted to write them logically to the best of my abilities so that they follow a structured narrative.

We begin with the exploration of multivariate analysis. When we deal with a multivariate function in $\mathbb{R}$, we deal with some map of the form $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. We can define this more formally via vector operations as the following:

$$f \left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right) = \begin{bmatrix} f_1(x_1, x_2, \dots, x_n) \\ \vdots \\ f_m(x_1, x_2, \dots, x_n) \end{bmatrix} \quad (1)$$

such that $\forall j \in [1, m] : f_j : \mathbb{R}^n \rightarrow \mathbb{R}$. Our original definition of the derivative in $\mathbb{R}$ was:

$$f'(x) := \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

or equivalently

$$\lim_{z \rightarrow x} \frac{f(z) - f(x) - f'(x)(z - x)}{z - x} = 0$$

Using the substitution $h = z - x$, we also get the equivalent definition:

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x) - f'(x)h}{h} = 0$$

In which case, our $\epsilon - \delta$ definition would be:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{|f(x + h) - f(x) - f'(x)h|}{|h|} < \epsilon$$

However if we try to plainly substitute the definition for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ into the first definition, we get "division of two vectors" which is not a defined operation for vector spaces (they are only abelian groups under addition).

We consider the following lemma in order to understand how to extend the derivative to the multivariate case:

Lemma 1.1. $f'(x)$ exists $\iff \left\{ \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| \right\}_{\delta > 0} \rightarrow 0$

Proof. $\implies$: If $f'(x)$ exists, then:

$$\begin{aligned} \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0, \forall z \in B(x, \delta) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot |z - x| < \epsilon \cdot \delta \end{aligned}$$

Because this is true for all z in the punctured δ -ball around x , we have an upper bound, so we take the supremum:

$$\implies \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta$$

And so we rearrange the terms to get a δ -net that converges such that:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0 : \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon$$

Thus proving the forward direction.

$\Leftarrow$: Now for the converse, consider:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0 : \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta$$

Now for any arbitrary δ -ball around x where $\delta \leq \delta_0$, we have that:

$$\begin{aligned} & \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta \\ \implies & \forall z \in B(x, \delta) \setminus \{x\} : \frac{1}{\delta} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \\ \implies & \forall z \in B(x, \delta_0) \setminus B(x, \delta) : \frac{|f(z) - f(x) - f'(x)(z - x)|}{|z - x|} < \frac{1}{\delta} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \end{aligned}$$

Since δ was arbitrary as long as it was less than or equal to δ_0 , we observe:

$$\begin{aligned} & \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0, \forall z \in B(x, \delta_0) \setminus B(x, \delta) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \iff & \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in \bigcup_{\delta \in (0, \delta_0]} (B(x, \delta_0) \setminus B(x, \delta)) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \iff & \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \bigcap_{\delta \in (0, \delta_0]} B(x, \delta) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \end{aligned}$$

Now because $\mathbb{R}$ is a T_1 space, we see that the intersection in the previous statement is equivalent to $\{x\}$, so we get:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon$$

□

We can define the function:

$$r(z) = \frac{f(z) - f(x) - f'(x)(z - x)}{z - x}$$

or $r(h)$ if we say $z = x + h$. Then taking limit as $z \rightarrow x$ (or $h \rightarrow 0$), we have that $r(z) \rightarrow 0$. Then the idea of the linear approximation is that:

$$f(z) = f(x) + f'(x)(z - x) + r(z)(z - x)$$

and the lemma above tells us that the worst remainder (similar to oscillation, which we called the "worst case net") converges to 0.

Definition 1.1. Given some $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ in the form of equation (1), then for some $i \in [1, m], j \in [1, n]$:

$$\frac{\partial f_i(x)}{\partial e_j} := \lim_{h \rightarrow 0} \frac{f_i(x + e_j h) - f_i(x)}{h}$$

where $\{e_1, \dots, e_n\}$ is the canonical basis of $\mathbb{R}^n$. This is the definition of the partial derivative.

Since each $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, we see that the partial derivative is scalar valued. We now show that it admits the linear approximation property. We can rewrite our limit definition as:

$$\lim_{h \rightarrow 0} \left(\frac{f_i(x + e_j h) - f_i(x)}{h} - \frac{\partial f_i(x)}{\partial e_j} \right) = \lim_{h \rightarrow 0} \frac{f_i(x + e_j h) - f_i(x) - \left(\frac{\partial f_i(x)}{\partial e_j} \right) h}{h} = 0$$

So with an argument similar to Lemma 1.1, we see that $f_i(z) \approx f_i(x) + \left(\frac{\partial f_i}{\partial e_j} \right) h$. Now observe that:

$$f(z) := \begin{bmatrix} f_1(z) \\ \vdots \\ f_m(z) \end{bmatrix}$$

Lemma 1.2. For all $i \in [1, m]$ and some $x \in \mathbb{R}^n$ and an e_j :

$$\frac{\partial f_i(x)}{\partial e_j} \text{ exist}$$

if and only if:

$$\frac{\partial f(x)}{\partial e_j} \text{ exists}$$

Proof. $\implies$: We claim that the partial derivative of f takes the following form:

$$\frac{\partial f(x)}{\partial e_j} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_j} \\ \vdots \\ \frac{\partial f_m(x)}{\partial e_j} \end{bmatrix}$$

We have some δ_i given by the existence of each partial derivative, so we define $\delta_0 := \min\{\delta_i : 1 \leq i \leq n\}$. Then for all $h \in B(0, \delta_0)$:

$$\frac{\|f(x + e_j h) - f(x) - \left(\frac{\partial f(x)}{\partial e_j} \right) h\|}{|h|} \leq \frac{1}{|h|} \sum_{i=1}^m \left| f_i(x + e_j h) - f_i(x) - \left(\frac{\partial f_i(x)}{\partial e_j} \right) h \right| < m\epsilon$$

where we used the inequality that the $\|v\| \leq \sum |v_i|$.

$\Leftarrow$: We claim that:

$$\frac{\partial f_i(x)}{\partial e_j} = e_i \cdot \frac{\partial f(x)}{\partial e_j}$$

We now use the inequality $|v_i| \leq \|v\|$ and find that $\forall i \in [1, m]$:

$$\frac{|f_i(x + e_j h) - f_i(x) - (e_i \cdot \frac{\partial f(x)}{\partial e_j}) h|}{|h|} = \frac{|e_i \cdot f(x + e_j h) - e_i \cdot f_i(x) - (e_i \cdot \frac{\partial f(x)}{\partial e_j}) h|}{|h|}$$

$$= \frac{|e_i \cdot (f(x + e_j h) - f(x) - \frac{\partial f(x)}{\partial e_j} h)|}{|h|} \leq \frac{\|f(x + e_j h) - f(x) - (\frac{\partial f(x)}{\partial e_j} h)\|}{|h|} < \epsilon$$

so the same δ suffices. □

Lemma 1.3. *The partial derivative of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ exists for some e_j if and only if there exists some $r(h) : \mathbb{R} \rightarrow \mathcal{M}_{m \times 1}(\mathbb{R})$ such that:*

$$f(x + h e_j) = f(x) + \frac{\partial f(x)}{\partial e_j} h + r(h)h \wedge \lim_{h \rightarrow 0} r(h) = 0$$

Proof. $\implies$: By the previous Lemma 1.2 and the results of Lemma 1.1, we can say that there exists some $r_i(h)$ for each f_i , such that:

$$r_i(h) = \frac{f_i(x + h e_j) - f_i(x) - \frac{\partial f_i(x)}{\partial e_j} h}{h}$$

Then, we can define $r(h)$ as:

$$r(h) = \begin{bmatrix} r_1(h) \\ \vdots \\ r_m(h) \end{bmatrix}$$

which tends to 0 by definition, or by the fact that $\|v\| \leq \sum |v_i|$.

$\Leftarrow$: We can rewrite the expression as:

$$r(h) = \frac{f(x + h e_j) - f(x) - \frac{\partial f(x)}{\partial e_j} h}{h}$$

so $\|r(h)\| \rightarrow 0$ implies that the quotient also tends to zero. □

Notice that the proofs did not use any properties of the basis vector e_j itself, so we can extend the definition:

Definition 1.2. *Given some $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ in the form of equation (1), then for some $v \in \mathbb{R}^n$ and $i \in [1, m], j \in [1, n]$:*

$$\frac{\partial f_i(x)}{\partial v} := \lim_{h \rightarrow 0} \frac{f_i(x + h v) - f_i(x)}{h}$$

and similarly for f . Such a derivative is called the Gateaux derivative.

We can extend the Gateaux derivative to obtain the Frechet derivative. The Gateaux derivative limited itself to a specific direction of the vector v (and then used h as the variable for the limit).

Definition 1.3. *The Frechet derivative of a function f at $x \in \mathbb{R}^n$ is:*

$$Df(x) : \lim_{v \rightarrow 0} \frac{\|f(x + v) - f(x) - (Df(x))(v)\|}{\|v\|} = 0$$

or equivalently:

$$\forall \epsilon > 0, \exists \delta > 0, \forall v \in B(0, \delta) : \frac{\|f(x + v) - f(x) - (Df(x))(v)\|}{\|v\|} < \epsilon$$

where the norm induces a metric so that $v \in B(0, \delta)$ (in this case 0 is the 0 vector), is the same as saying $\|v\| \in B(0, \delta)$ or equivalently $\|v\| < \delta$.

The form of the limit is similar to the idea of linear approximation. Since $v \in \mathbb{R}^n$, then we must have that $Df(x) \in \mathcal{M}_{m \times n}(\mathbb{R})$. If the function is Frechet differentiable, then $Df(x)$ is the Jacobian:

$$Df(x) = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_1} & \dots & \frac{\partial f_1(x)}{\partial e_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(x)}{\partial e_1} & \dots & \frac{\partial f_m(x)}{\partial e_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f(x)}{\partial e_1} & \dots & \frac{\partial f(x)}{\partial e_n} \end{bmatrix}$$

Proof. If we chose the family of vectors $\{he_j : h \in \mathbb{R} \wedge j \in [1, n]\}$, then differentiability tells us:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{\|f(x + he_j) - f(x) - h(Df(x))e_j\|}{|h|} < \epsilon$$

telling us that:

$$\frac{\partial f(x)}{\partial e_j} = Df(x)e_j$$

which is the j -th column of the Jacobian matrix. □

Notice that Lemma 1.2 was a statement on equivalence; however, the converse is not necessarily true for the Jacobian matrix. The total differentiability of f at x gives us the existence of partial derivatives, but not necessarily the converse. In order to prove this, we require the following concept:

1.1.1 MVT in the Multivariable Case

Theorem 1.1. *Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous on the closed interval $[a, b]$ and differentiable on (a, b) , then:*

$$\exists z \in (a, b) : f'(z) = \frac{f(b) - f(a)}{b - a}$$

Proof. Since f is continuous on a compact interval, it achieves the extrema by EVT. Now consider the function:

$$h(x) = \frac{f(b) - f(a)}{b - a}(x - a) - f(x) + f(a)$$

which at both $x = a$ and $x = b$, we have that $h(x) = 0$, so by Rolle's Theorem, there exists some extrema within the interval, so that:

$$\exists z \in (a, b) : h'(z) = 0 \implies f'(z) = \frac{f(b) - f(a)}{b - a}$$

□

If we try to apply this proof to a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, then in order to apply Theorem 1.1, we require some compact interval given f is continuous.

Now, the following lemma will be useful in exploring MVT:

Lemma 1.4. *Any open, convex subset $U \subseteq \mathbb{R}^n$ is path connected.*

Proof. Openness does not play an important role here. We can simply take the continuous function:

$$f(\alpha) = \alpha x + (1 - \alpha)y$$

for some fixed $x, y \in U$. By convexity, we know $\text{Ran}(f) \subseteq U$. □

Lemma 1.5. *An open ball $B(x, r)$ is convex.*

Proof. Take $y, z \in B(x, r)$, then:

$$|\alpha y + (1 - \alpha)z - x| = |\alpha(y - x) + (1 - \alpha)(z - x)| \leq \alpha r + (1 - \alpha)r = r$$

□

Theorem 1.2. *Given $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, we have that for some open, convex subset $U \subseteq \mathbb{R}^n$ (on which f_i is differentiable) and given two points a, b :*

$$\exists z \in U : Df_i(z)(b - a) = f_i(b) - f_i(a)$$

where $Df_i(z)$ is the gradient vector at z .

Proof. Since U is convex, we can take the function for $t \in [0, 1]$:

$$h(t) = tb + (1 - t)a$$

such that $h(0) = a$ and $h(1) = b$. Then applying Theorem 1.1 to the function $f_i \circ h$ (which too is continuous on $[0, 1]$ and differentiable on $(0, 1)$), we see that:

$$\exists c \in (0, 1) : (f_i \circ h)'(c) = \frac{f_i(h(1)) - f_i(h(0))}{1 - 0} = f_i(b) - f_i(a)$$

and $(f_i \circ h)'(c) = Df_i(h(c)) \cdot h'(c) = Df_i(c)(b - a)$, where:

$$Df_i(x) = \left[\frac{\partial f_i(x)}{\partial e_1} \quad \dots \quad \frac{\partial f_i(x)}{\partial e_n} \right]$$

We used the multivariate chain rule in this proof, where $(f \circ h)'(t) = Df(h(t))Dh(t)$. In this case, we defined:

$$h(t) = \begin{bmatrix} tb_1 + (1 - t)a_1 \\ \vdots \\ tb_n + (1 - t)a_n \end{bmatrix}$$

and since each $h_i : \mathbb{R} \rightarrow \mathbb{R}$, then $Dh(t)$ is an $n \times 1$ matrix in the e_1/t direction, the form of which is given by Lemma 1.2:

$$Dh(t) = \begin{bmatrix} b_1 - a_1 \\ \vdots \\ b_n - a_n \end{bmatrix} = b - a$$

□

Corollary 1.1. *If $\exists k \in \mathbb{R}, \exists j \in [1, n] : b - a = ke_j$ and the partial derivative of f_i exists in the e_j direction, then:*

$$\exists z \in \text{span}(e_j) : \frac{\partial f_i(z)}{\partial e_j} k = f_i(b) - f_i(a)$$

Notice this corollary does not require full differentiability of f_i (which means it does not require the gradient to be the total derivative of f at said point). We can expand MVT to the vector valued function case with an inequality.

Corollary 1.2. Given $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and an open, convex subset $U \subseteq \mathbb{R}^n$ on which f is totally differentiable and $\|Df(x)\|$ is bounded, then:

$$\forall a, b \in U : \|f(b) - f(a)\| \leq M\|b - a\|$$

such that $\exists M \in \mathbb{R}, \forall x \in U : \|Df(x)\| \leq M$.

Proof. Consider any arbitrary $v \in \mathbb{R}^n$, then:

$$\forall i \in [1, m] : \frac{\|Df_i(v)v\|}{\|v\|} \leq \frac{\|Df(v)v\|}{\|v\|} \leq M$$

Thus $\|Df_i(v)\| \leq M$, so we can apply Theorem 1.2 and see the following:

$$\|f(b) - f(a)\| \leq \sum_{i=1}^m \|f_i(b) - f_i(a)\| \leq \sum_{i=1}^m \|Df_i(z_i)\| \cdot \|b - a\| \leq mM\|b - a\|$$

This is not the estimate we were looking for, but one possible estimate. We can derive another by the following:

$$\begin{aligned} \|f(b) - f(a)\| &= \sqrt{\sum_{i=1}^m |f_i(b) - f_i(a)|^2} = \sqrt{\sum_{i=1}^m |Df_i(z_i)(b - a)|^2} \\ &\leq \sqrt{\sum_{i=1}^m C^2 \|b - a\|^2} = \sqrt{m} \cdot C \|b - a\| \end{aligned}$$

Consider once again the intermediate function $h(t)$ where $h : [0, 1] \rightarrow \mathbb{R}^n$ introduced in our proof of Theorem 1.2, then:

$$g(t) = (f(b) - f(a))^T f(h(t)) \implies g'(t) = (f(b) - f(a))^T Df(h(t))(b - a)$$

thus $g'(t) : \mathbb{R} \rightarrow \mathbb{R}$. Then define $g : [0, 1] \rightarrow \mathbb{R}$:

$$|g'(t)| \leq \|f(b) - f(a)\| \cdot \|Df(h(t))\| \cdot \|b - a\| \leq \|f(b) - f(a)\| \cdot M\|b - a\|$$

We can apply regular MVT to g , which tells us:

$$|g(1) - g(0)| = |(f(b) - f(a))^T (f(b) - f(a))| = \|f(b) - f(a)\|^2 \leq \|f(b) - f(a)\| \cdot M\|b - a\|$$

and the intended inequality follows. $\square$

Lemma 1.6. The function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is totally differentiable at a point if:

1. The partial derivatives exist
2. The partial derivatives are continuous at that point

Proof. We want to show:

$$\forall \epsilon > 0, \exists \delta > 0, \forall v \in B(0, \delta) : \frac{\|f(x+v) - f(x) - (Df(x))v\|}{\|v\|} < \epsilon$$

The expression $(Df(x))(v)$ gives us the following vector:

$$(Df(x))(v) = \begin{bmatrix} \sum_{i=1}^n \frac{\partial f_1(x)}{\partial e_i} v_i \\ \vdots \\ \sum_{i=1}^n \frac{\partial f_m(x)}{\partial e_i} v_i \end{bmatrix}$$

and we see that:

$$\frac{\|f(x+v) - f(x) - (Df(x))(v)\|}{\|v\|} \leq \frac{1}{\|v\|} \sum_{j=1}^m \left| f_j(x+v) - f_j(x) - \sum_{i=1}^n \frac{\partial f_j(x)}{\partial e_i} v_i \right|$$

Continuity of the partial derivatives give us:

$$\forall \epsilon > 0, \exists \delta_{ji} > 0, \forall v \in B(0, \delta_{ji}) : \left| \frac{\partial f_j(x+v)}{\partial e_i} - \frac{\partial f_j(x)}{\partial e_i} \right| < \epsilon$$

We set $\delta := \min\{\delta_{ji} : 1 \leq j \leq m \wedge 1 \leq i \leq n\}$. Then given some $v \in B(0, \delta)$, we apply the idea in Corollary 1.1 and see that:

$$f_j(x+v) - f_j(x) = \sum_{i=1}^n f_j(x+V_i) - f_j(x+V_{i-1})$$

where we define $V_i := v_1 e_1 + \dots + v_i e_i$. Then, we can repeatedly apply Corollary 1.1 and get:

$$= \frac{1}{\|v\|} \sum_{j=1}^m \left| \sum_{i=1}^n \left(\frac{\partial f_j(z_i)}{\partial e_i} - \frac{\partial f_j(x)}{\partial e_i} \right) v_i \right| \leq \frac{\epsilon \cdot mn \cdot \|v\|}{\|v\|} = \epsilon \cdot mn$$

□

The MVT introduced in Theorem 1.2 fails for vector valued functions as we cannot guarantee the same z satisfies the equality for all f_i ; however, there are some generalizations to this case which we can expound upon later. Finally, we can expand the idea of linear approximation to the total vector valued function:

Lemma 1.7. $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is totally differentiable if and only if there exists $r(v) : \mathbb{R}^n \rightarrow \mathcal{M}_{m \times n}(\mathbb{R})$ such that:

$$f(x+v) = f(x) + Df(x)(v) + r(v)v \wedge \lim_{v \rightarrow 0} \|r(v)\| = 0$$

Proof. Observe that we use the matrix norm, which is defined as:

$$\begin{aligned} \|A\| &= \sup_{x \in \mathbb{R}^n} \frac{\|Ax\|}{\|x\|} \\ \forall x \in \mathbb{R}^n : \frac{\|Ax\|}{\|x\|} &= \sqrt{\frac{\sum_{i=1}^m (\sum_{j=1}^n A_{ij} x_j)^2}{\sum_{j=1}^n x_j^2}} \\ &\leq \sqrt{\frac{\sum_{i=1}^m (\sum_{j=1}^n A_{ij}^2 \sum_{j=1}^n x_j^2)}{\sum_{j=1}^n x_j^2}} = \sqrt{\sum_{i=1}^m \left(\sum_{j=1}^n A_{ij}^2 \right)} \end{aligned}$$

and since this is true for all $x \in \mathbb{R}^n$, we have a concrete upper bound, so:

$$\|A\| \leq \sqrt{\sum_{i=1}^m \left(\sum_{j=1}^n A_{ij}^2 \right)}$$

Now, we can use the fact that $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ and that its total derivative exists as seen by:

$$\frac{|f_i(x+v) - f_i(x) - Df_i(x)(v)|}{\|v\|} = \frac{|e_i^T (f(x+v) - f(x) - Df(x)(v))|}{\|v\|} \leq \frac{\|f(x+v) - f(x) - Df(x)(v)\|}{\|v\|}$$

Then, we say that equivalently there exists some $r_i(v) : \mathbb{R}^n \rightarrow \mathcal{M}_{1 \times n}$ such that:

$$r_i(v)v = f_i(x+v) - f_i(x) - Df_i(x)(v)$$

such that:

$$r_i(v) = \left[(f_i(x+v) - f_i(x) - Df_i(x)(v)) \frac{v_1}{\|v\|^2} \quad \cdots \quad (f_i(x+v) - f_i(x) - Df_i(x)(v)) \frac{v_n}{\|v\|^2} \right]$$

$$\implies \|r_i(v)\| = \|f_i(x+v) - f_i(x) - Df_i(x)(v)\| \sqrt{\frac{1}{\|v\|^2}} = \frac{\|f_i(x+v) - f_i(x) - Df_i(x)(v)\|}{\|v\|} \rightarrow 0$$

The converse follows from the fact that $\|r_i(v)v\| \leq \|r_i(v)\| \cdot \|v\|$. Now, we can define:

$$r(v) := \begin{bmatrix} r_1(v) \\ \vdots \\ r_m(v) \end{bmatrix}$$

for which by the inequality for matrix norm we derived above, we have that:

$$\|r(v)\| \leq \sqrt{\sum_{i=1}^m (f_i(x+v) - f_i(x) - Df_i(x)(v))^2} \cdot \frac{1}{\|v\|^2} = \frac{1}{\|v\|} \cdot \|f(x+v) - f(x) - Df(x)(v)\| \rightarrow 0$$

This is a tighter ϵ -bound than using the fact (which we can prove) that:

$$\|r(v)\| \leq \sum_{i=1}^m \|r_i(v)\| < m\epsilon$$

taking δ minimum (follows from the fact that if f is differentiable, then f_i is differentiable). The converse follows from the fact that $\|r(v)v\| \leq \|r(v)\| \cdot \|v\|$. $\square$

Now we can prove the following:

Lemma 1.8. *If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at x , then it is continuous at x .*

Proof. By Lemma 1.7, we have that there exists some $r(v) : \mathbb{R}^n \rightarrow \mathcal{M}_{m \times n}(\mathbb{R})$ such that:

$$f(x+v) = f(x) + Df(x)(v) + r(v)v$$

Now, let $\|r(v)\| < 1$, which gives us some δ_0 such that for all $v \in B(0, \delta_0)$:

$$\begin{aligned} \|f(x+v) - f(x)\| &= \|(Df(x) + r(v))v\| \leq \|(Df(x) + r(v))\| \cdot \|v\| \\ &\leq \|v\| \cdot (\|Df(x)\| + 1) \end{aligned}$$

So taking:

$$\delta := \min \left\{ \delta_0, \frac{\epsilon}{1 + \|Df(x)\|} \right\}$$

we see that:

$$\forall v \in B(0, \delta) : \|f(x+v) - f(x)\| \leq \|v\| \cdot (\|Df(x)\| + 1) \leq \epsilon$$

$\square$

1.2 Inverse and Implicit Function Theorems

The idea of the inverse function theorem stems from our idea of the linear approximation where $f(x+v) \approx f(x) + Df(x)v$. Then $v \approx Df(x)^{-1}(f(x+v) - f(x))$.

Theorem 1.3 (Banach Fixed Point Theorem). *Given a contraction map $f : X \rightarrow X$ and (X, ϱ) a complete metric space, such that $\varrho(f(x), f(y)) \leq c\varrho(x, y)$, where $c \in [0, 1)$, then there exists a unique fixed point such that:*

$$\exists x' \in X : f(x') = x'$$

Proof. Let us choose an arbitrary point $x \in X$. Then:

$$\varrho(f(x), f(f(x))) \leq c\varrho(x, f(x))$$

Then, we see that:

$$\varrho(f^n(x), f^{n+1}(x)) \leq c^n \varrho(x, f(x))$$

So for any arbitrary $m \leq n$:

$$\begin{aligned} \varrho(f^m(x), f^n(x)) &\leq \sum_{i=0}^{n-m-1} \varrho(f^{m+i}(x), f^{m+i+1}(x)) \leq \varrho(x, f(x)) \sum_{i=0}^{n-m-1} c^{m+i} \\ &= \varrho(x, f(x)) \sum_{i=m}^{n-1} c^i = \frac{c^m(1 - c^{n-m+1})}{1 - c} \cdot \varrho(x, f(x)) \end{aligned}$$

We now show that the sequence $\{f^{(n)}(x)\}_{n \in \mathbb{N}}$ is Cauchy:

$$\frac{c^m(1 - c^{n-m+1})}{1 - c} \cdot \varrho(x, f(x)) \leq \frac{c^m}{1 - c} \cdot \varrho(x, f(x)) < \epsilon$$

So, if we choose n_0 so that:

$$\frac{c^{n_0}}{1 - c} \cdot \varrho(x, f(x)) < \epsilon$$

then for all $m, n \geq n_0$, we have that the sequence is Cauchy. Because X is complete, we have that there exists some $x' \in X$ such that:

$$f^n(x) \rightarrow x'$$

Now, f is c -Lipschitz, so by continuity, we have that the sequence $\{f^{n+1}(x)\}_{n \in \mathbb{N}}$ converges to $f(x')$. Since it is a subsequence of our initial sequence, they must converge to the same limit, thus $f(x') = x'$.

The fixed point is also guaranteed to be unique by the uniqueness of a limit in a T_2 space, which we shall prove. Let us assume there exists another fixed point $\tilde{x}$ such that $f(\tilde{x}) = \tilde{x}$. Then:

$$\varrho(f(\tilde{x}), f(x')) = \varrho(\tilde{x}, x') \leq c\varrho(\tilde{x}, x') < \varrho(\tilde{x}, x')$$

which is a contradiction. However, if we allowed $c = 1$, then we may not have a unique fixed point (the Cauchy estimate may also fail). $\square$

Theorem 1.4. *If $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable on an open ball $B(x_0, \delta_0)$ with $Df(x_0)$ continuous and invertible, then there exists some $B(x_0, \delta) \subseteq B(x_0, \delta_0)$, then:*

1. f is injective on $B(x_0, \delta)$ and an open map
2. f^{-1} is differentiable on $f(B(x_0, \delta))$
3. $D(f^{-1})$ is continuous at $f(x_0)$

Proof. Part One

We are given that $\forall x \in B(x_0, \delta) : \|Df(x) - Df(x_0)\| < \epsilon$. We want to show that:

$$\forall x, y \in B(x_0, \delta) : f(x) = f(y) \implies x = y$$

We can define $\delta := \min\{\delta_0/2, \delta_1\}$ where δ_1 is given by continuity so that $B(x_0, \delta) \subseteq B(x_0, \delta_0)$ for $\epsilon = \frac{1}{2(\|Df^{-1}(x_0)\|+1)}$, then consider the function:

$$h(x) = x + Df(x_0)^{-1}(y - f(x))$$

for some arbitrary $y \in \text{Ran}(f|_{B(x_0, \delta)})$. We show that h is a contraction map:

$$Dh(x) = I - Df^{-1}(x_0)Df(x) = Df(x_0)^{-1}(Df(x_0) - Df(x))$$

$$\|Dh(x)\| \leq \|Df(x_0)^{-1}\| \cdot \|Df(x_0) - Df(x)\| \leq \frac{1}{2}$$

and then by Corollary 1.2, we have that:

$$\forall x, y \in B(x_0, \delta) : \|h(x) - h(y)\| \leq \frac{1}{2}\|x - y\|$$

so it is a contraction map, thus any fixed points are unique, giving us that f is injective (and furthermore trivially surjective). Notice that $B(x_0, \delta)$ is not closed (thus not complete), so we cannot apply fixed point theorem. However, we do not require fixed point theorem since the fixed points are guaranteed as $f(B(x_0, \delta))$ is the image of the open ball.

Part One (Alternate Proof)

To show injectivity, it suffices to show that there exists some $c \in \mathbb{R}$ such that:

$$\forall x_1, x_2 \in B(x_0, \delta) : \|x_1 - x_2\| \leq c\|f(x_1) - f(x_2)\|$$

We consider the equivalent criterion of differentiability whereby:

$$f(x_0 + v) = f(x_0) + Df(x_0)(v) + r(v)v \wedge \|r(v)\| \rightarrow 0$$

We can define $R(v) = r(v)v$, for which we can prove the equivalent condition:

$$f(x_0 + v) = f(x_0) + Df(x_0)(v) + R(v) \wedge \frac{\|R(v)\|}{\|v\|} \rightarrow 0$$

So, we see that $DR(v) = Df(x_0 + v) - Df(x_0)$, so by continuity, we see that, taking the same ϵ as before:

$$\forall v \in B(0, \delta) : \|DR(v)\| = \|Df(x_0 + v) - Df(x_0)\| \leq \frac{1}{2\|Df(x_0)^{-1}\|}$$

though, an arbitrary ϵ of say $1/2$ would suffice. Thus by Corollary 1.2, since $B(x_0, \delta)$ is open and convex:

$$\begin{aligned} \forall v, w \in B(0, \delta) : \|R(v) - R(w)\| &\leq \frac{\|Df(x_0)\|}{2}\|v - w\| \\ R(v) - R(w) &= f(x_0 + v) - f(x_0) - Df(x_0)(v) - f(x_0 + w) + f(x_0) + Df(x_0)(w) \\ \|R(v) - R(w)\| &= \|f(x_0 + v) - f(x_0 + w) - Df(x_0)(v - w)\| \\ \implies \|Df(x_0)(v - w)\| - \|f(x_0 + v) - f(x_0 + w)\| &\leq \frac{1}{2\|Df(x_0)^{-1}\|}\|v - w\| \end{aligned}$$

We also know that $\|x\| = \|Df(x_0) \cdot Df(x_0)^{-1}x\| \leq \|Df(x_0)^{-1}\| \cdot \|Df(x_0)x\|$, thus

$$\implies \frac{1}{2\|Df(x_0)^{-1}\|}\|v - w\| \leq \|f(x_0 + v) - f(x_0 + w)\|$$

and a simple algebraic rearrangement brings us to the desired result.

The idea of this contraction map and the alternative proof both come from our idea of linear approximation, where:

$$f(z) \approx f(x) + Df(x)(z - x)$$

Part Two

We now prove that f is an open map, so we want to show that $f(B(x_0, \delta))$ is open:

$$\forall x \in B(x_0, \delta), \exists \delta' > 0 : B(f(x), \delta') \subseteq f(B(x_0, \delta))$$

Then consider some $\tilde{\delta}$ such that $\overline{B(x, \tilde{\delta})} \subseteq B(x_0, \delta)$. Then we claim that the ball $\epsilon\tilde{\delta}$ is the δ' we are looking for. We use the same idea of a contraction map, but this time apply fixed point theorem. The reason we use fixed point theorem is we want to show that:

$$\forall y \in B(f(x), \epsilon\tilde{\delta}), \exists z \in \overline{B(x, \tilde{\delta})} : f(z) = y$$

telling us that $B(f(x), \epsilon\tilde{\delta}) \subseteq f(\overline{B(x, \tilde{\delta})}) \subseteq f(B(x_0, \delta))$. So for a fixed $y \in B(f(x), \epsilon\tilde{\delta})$:

$$\begin{aligned} \|h(z) - x\| &\leq \|h(z) - h(x)\| + \|h(x) - x\| \leq \frac{1}{2}\|z - x\| + \|Df(x_0)^{-1}\| \cdot \|y - f(x)\| \\ &< \frac{\tilde{\delta}}{2} + \|Df(x_0)^{-1}\| \epsilon\tilde{\delta} = \tilde{\delta} \end{aligned}$$

so h which we showed is a contraction map, is a contraction map on $\overline{B(x, \tilde{\delta})}$, thus by Banach Fixed Point theorem, for each arbitrary y , there exists a fixed point (note we need the theorem because we are not immediately guaranteed the existence of some z such that $f(z) = y$).

Part Three

We now need to show the existence of $Df^{-1}(x)$ for $x \in B(x_0, \delta)$. A matrix is invertible if and only if all of its eigenvalues are nonzero. Thus:

$$\forall i \in [1, n] : Df(x)v_i = \lambda_i v_i \wedge \lambda_i \neq 0$$

We can rewrite $Df(x)$ as $Df(x_0) + (Df(x) - Df(x_0))$, so we see that:

$$\begin{aligned} \|Df(x_0)x + (Df(x) - Df(x_0))x\| &= \|Df(x_0)(x + Df(x_0)^{-1}(Df(x) - Df(x_0))x)\| \\ &\geq \frac{1}{\|Df(x_0)^{-1}\|} (\|x\| - \|Df(x_0)^{-1}\| \cdot \|Df(x) - Df(x_0)\| \cdot \|x\|) \\ &\geq \frac{\|x\|}{\|Df(x_0)^{-1}\|} (1 - \|Df(x_0)^{-1}\| \cdot \|Df(x) - Df(x_0)\|) \geq \frac{\|x\|}{2\|Df(x_0)^{-1}\|} \end{aligned}$$

Now assume $Df(x)$ is non-invertible, which means it has some v_i with $\lambda_i = 0$, then:

$$0 \leq \frac{\|v_i\|}{2\|Df(x_0)^{-1}\|} \leq 0$$

which forces $v_i = 0$, which is a contradiction as an eigenvector cannot be the zero vector. Thus $Df(x)$ is invertible for $x \in B(x_0, \delta)$ (observe it is only true for this ball as continuity gives us a strict lower bound of 0). Otherwise, it may be that:

$$\|Df(x_0)^{-1}\| \cdot \|Df(x) - Df(x_0)\| > 1$$

in which case the argument does not hold, since we do not get a squeeze inequality. We now must show that $Df^{-1}(f(x)) = Df(x)^{-1}$, given the invertibility of the derivative (which we just proved), injectivity of the function (proved in part one), and continuity of the inverse (also proved in part one).

□

1.3 Configuration Spaces

The idea of configuration spaces is intertwined with the idea of manifolds, which I have not studied, so for this section we will limit ourselves to Euclidean configuration spaces.